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Deep Compressed Sensing based Data Imputation for Urban Environmental Monitoring

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Research output: Contribution to journal › Article › peer-review

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Abstract

Data imputation is prevalent in crowdsensing, especially for Internet of Things (IoT) devices. On the one hand, data collected from sensors will inevitably be affected or damaged by unpredictability. On the other hand, extending the active time of sensor networks has urgently aspired environmental monitoring. Using neural networks to design a data imputation algorithm can take advantage of the prior information stored in the models. This paper proposes a preprocessing algorithm to extract a

subset for training a neural network on an IoT dataset, including time window determination, sensor aggregation, sensor exclusion and data frame shape selection. Moreover, we propose a data imputation algorithm using deep compressed sensing with generative models. It explores novel representation matrices and can impute data in the case of a high missing ratio situation. Finally, we test our subset extraction algorithm and data imputation algorithm on the EPFL SensorScope dataset, respectively, and they effectively improve the accuracy and robustness even with extreme data loss.

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Keywords

- Compressed sensing
- crowdsensing
- generative models
- data imputation

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KW - crowdsensing

KW - generative models

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